Tutorial 7– Acid/base Titration

T7-1. 100 mL of acetic acid (CH_3COOH) with a concentration of 0.05M (pKa = 4.76) was titrated with a 0.1M solution of NaOH. Calculate the pH at:

a. Before adding the base.

b. At 20% of the titration.

c. At 50% of the titration.

d. At 99% of the titration.

e. At the equivalent point

f. 0.5 mL excess of NaOH after the equivalent point.

g. 10 mL excess of NaOH after the equivalence point Make a graph of pH as a function of the volume of the base that was added.

T7-2. 50 mL of the base methylamine (CH_3NH_2) with a concentration of 0.042M was titrated with HCl and this are the results:

V(HCl)	pH	V(HCl)	pH	V(HCl)	pH
0	11.959	10	10.686	20	9.344
1	11.946	11	10.604	21	6.087
2	11.623	12	10.52	22	2.556
3	11.423	13	10.434	23	2.386
4	11.273	14	10.344	24	2.267
5	11.15	15	10.247	25	2.176
6	11.043	16	10.14	26	2.103
7	10.946	17	10.017	27	2.041
8	10.856	18	9.867	28	1.989
9	10.77	19	9.667	29	1.943

Given that the reaction is: $CH_3NH_2 + HCl \longrightarrow CH_3NH_3^+ + Cl^-$

a. What is the K_a of $CH_3NH_3^+$ (What is the K_b for the methylamine CH_3NH_2)?

b. What is the concentration of the HCl?

T7-3. Calculate the titration error if we stop the titration of V_0 = 50.0 mL of a C_0 = 0.10 M solution of HA with Ka = 5×10^{-5} at end point pH = 7.52 instead of the theoretical pH= 8.50. The titration is performed with a C = 0.10 M solution of NaOH.